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29 January 1988
SUPERSEDING
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MILITARY SPECIFICATION

PROPELLANT M9 FLAKE, FOR USE IN IGNITION CARTRIDGES

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of M9 Flake propellant for use in ignition cartridges (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286 - Propellant, Solid, Sampling, Examination and Testing
MIL-STD-652 - Propellants, Solid, for Cannon, Requirements and Packing

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8837347 - Cartridge, Ignition, M66
8865062 - Cartridge, Ignition, M6 Assembly
8865089 - Cartridge, Ignition, M8 Assembly
8880647 - Cartridge, Ignition, M5A1 Assembly
9233373 - Cartridge, Ignition, M66A1 Assembly
9240960 - Cartridge, Ignition, M285, Loading Assembly
9242127 - Cartridge, Ignition, M5A2
9280553 - Cartridge, Ignition, M702, Loading Assembly
9293422 - Cartridge, Ignition, M299, Loading Assembly
9294752 - Cartridge, Ignition, M752A1, Loading Assembly

CODE OF FEDERAL REGULATIONS

Title 49 CFR - Transportation, Parts CFR 171-179.

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall

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supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. The propellant shall be M9 Flake conforming to the drawings of Table I and the requirements of MIL-STD-652. The chemical composition shall be determined in accordance with 4.5.1.

TABLE I. Drawing requirements.

Ignition Cartridge Designation	Drawing	Type Mortar
M5A1	8880647	60mm
M5A2	9242127	60mm
M6	8865062	81mm
M8	8865089	81mm
M66	8837347	81mm
M66A1	9233373	81mm
M285	9240960	81mm
M299	9293422	81mm
M702	9280553	60mm
M752A1	9294752	81mm

3.2 Physical properties. The propellant physical properties shall be as specified in Table II. The M9 diameter, web thickness, and bulk density shall be determined in accordance with 4.5.1.

TABLE II. Propellant physical properties.

Ignition Cartridge Designation	M9 Diameter, inches, (nominal)	M9 Web thickness inches, (nominal)	M9 Bulk density lbs/cu. ft. (minimum)
M5A1	0.035	0.00325	32.0
M5A2	0.035	0.00325	32.0
M6	0.059	0.010	46.0
M8	0.059	0.010	46.0
M66	0.059	0.010	46.0
M66A1	0.059	0.010	46.0
M285	0.059	0.010	46.0
M299	0.059	0.010	46.0
M702	0.035	0.006	32.0
M752A1	0.059	0.010	46.0

3.3 Proving ground assessment. The propellant when fired in the appropriate Mortar, shall comply with the requirements of Table III (some alternate combinations of ignition cartridges and ammunition may be used for assessment when approved by the technical agency, see 6.6) when tested as specified in 4.5.2 at charge O.

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TABLE III. Proving ground requirements.

Ignition cartridge designation	Mortar tube	Cartridge	Cartridge as fired (lbs) (+0.02)	Service vel. (ft/sec)	70°F Max. std. dev. (ft/sec)
M5A1	M19	M49A2	2.90	185	2.0
M5A2	M19	M49A4	3.10	170	2.0
M6	M1	M301A2	10.08	170	3.0
M8	M1	M43A1	7.04	240	2.5
M66	M29A1	M374A3	9.30	180	2.4
M66A1	M29A1	M301A3	10.20	162	3.0
		M374A2	9.12	210	2.4
M285	M29	M374A2	9.12	210	2.4
M299	M29	M374A3	9.05	215	2.4
M702	M224	M720	3.75	210	3.5
M752A1	M252	M819	10.35	210	3.1

3.4 Static pressure. The average internal ignition pressure plus 3 standard deviations shall not exceed 23,000 psi when determined in accordance with 4.5.3.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility of compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction Sample Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 Preproduction sample (pilot lots). The preproduction sample is intended to provide necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. The contractor shall select propellant granulations for approval by Government Technical Agency (see 6.6), which from experience are expected to bracket the requirements of this specification. The contractor shall submit a preproduction sample of sufficient quantity for all required tests, as designated by the Contracting Officer for evaluation in accordance with 4.3.2, usually consisting of up to three pilot lots.

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 shall be subjected by the government to any or all of the tests specified in this specification or applicable drawings (see MIL-A-48078).

4.3.3 Rejection. Government Engineering and Product Assurance personnel and the contractor shall review all pilot lot test data and shall select the granulation which best fits the requirements of this specification. The technical agency shall notify the contracting officer of the selected granulation. The contracting officer shall designate the chosen pilot lot as the accepted preproduction sample and shall authorize the contractor to produce the selected granulation. If none of the pilot lots submitted meets or brackets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. Until the preproduction sample is accepted, the contractor is not authorized by the government to initiate regular production, unless otherwise directed by the contracting officer. All future production lots should be of the same basic granulation as the accepted pilot lot. (See MIL-A-48078).

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4.4 Quality conformance inspection. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1 Examination and testing.
PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.1.1 Sampling for chemical and physical testing. Representative samples shall be taken by random selection of five containers from each propellant lot. Equivalent quantities from each of the five containers shall be taken such that the aggregate quantity shall be sufficient for the required tests. These samples shall be poured into clean containers, immediately closed with a hermetic seal and identified with a label showing the container number from which the sample was taken, the lot number, propellant nomenclature, and packing date. Prior to conducting lot acceptance tests, these samples shall be permitted to attain 70 $\pm$ 5°F; then equal portions sufficient to make the required quantity of a composite sample shall be taken from each representative sample and blended. The composite sample shall be placed in containers, immediately closed with a hermetic seal, and identified respectively as "Composite Sample". The "Composite Sample" shall be used for chemical and physical tests. If the sample fails to meet any of the requirements specified in section 3, the lot shall be rejected.

4.4.1.2 Chemical and physical characteristics.

4.4.1.2.1 Finished propellant. Each lot of propellant shall be tested for verification of composition and physical properties in accordance with 4.5.1.

4.4.1.3 Ballistic performance. Five containers shall be selected at random from each lot of propellant. Equivalent quantities of the five containers shall be taken such that the aggregate quantity shall be sufficient for ballistic performance tests described in 4.5.2.

4.4.1.3.1 Muzzle velocity standard deviation. If the velocity standard deviation of the rounds tested in accordance with 4.5.2.2 exceeds the applicable requirement, apply with retest criteria in accordance with 4.5.2.3. If the retest criteria cited does not apply, the lot shall be rejected.

4.4.1.4 Static pressure. The static pressure shall be tested as specified in 4.5.3. Ten (10) ignition cartridges shall be

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loaded at the recommended charge weight of M9 propellant. If the average ignition cartridge pressure exceeds the specified requirement, the lot shall be rejected.

4.5 Methods of inspection.

4.5.1 Chemical composition and physical properties (see 4.4.1.1). The propellant shall be tested for chemical composition and physical properties in accordance with MIL-STD-652. The bulk density shall be determined in accordance with method 502.1 or 502.2 of MIL-STD-286.

4.5.2 Proving ground assessments. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. The following temperature conditioning tolerances shall apply: -50 + 5°F, 70 ± 2.5°F, 145 -5°F.

4.5.2.1 Charge establishment.

4.5.2.1.1 Charge slope method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F, and an angle of fire of 45 degrees. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.3). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.2.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.5.2.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can then be used as in 4.5.2.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.5.2.2 Uniformity.

4.5.2.2.1 Initial production (see 6.7). Thirty rounds shall be assembled with the established charge weight of propellant.

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Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round) and fired at an angle of fire of 45°. Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard cartridge as fired weight (see 3.3) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F.

4.5.2.2.2 Subsequent production. Ten (10) rounds shall be assembled and tested at 70°F, fired alternately with calibration rounds at an angle of fire of 45°. The means and standard deviations of the velocity and pressure shall be calculated on results corrected to standard conditions and recommended charge weight.

4.5.2.3 Retest criteria. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Test lot fails velocity standard deviation or average pressure at 70°F and calibration rounds fail one or more requirements stated in 4.5.2.4, retest for uniformity.

b. Test lot fails velocity standard deviation or average pressure at 70°F and any extreme test value is an outlier, retest for uniformity.

c. The final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.5.2.1.1.

4.5.2.4 Referral criteria. A test lot shall be referred to U.S. Army Armament Research, Development and Engineering Center (ARDEC), Picatinny Arsenal, NJ 07806-5000, ATTN: AMSMC-QAR-R(D), for disposition whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

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b. Calibration correction applied exceeds two percent of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration rounds fired is less than specified.

e. Number of test or calibration results utilized in calculations is less than specified.

4.5.3 Static pressure test. (see 4.4.1.4) Determine the static pressure of the ignition cartridge by an accepted established method. The ignition cartridge should be assembled into the applicable production fin and tested at 70 ± 5 °F. The pressure should be tapped at the rear-most flash hole.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use in 60mm and 81mm mortar ignition cartridges.

6.2 Ordering data. See MIL-A-48078.

6.3 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description Sheet shall be forwarded to Commander, US Army Armament Research, Development, and Engineering Center (ARDEC), ATTN: AMSMC-QAR-R (D), and one copy to ATTN: SMCAR-AEE-BP, Picatinny Arsenal, NJ 07806-5000.

6.4 Equivalent test methods. The test methods given in this specification are the offical methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: ARDEC, ATTN: AMSMC-QAR-R (D), Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy, precision and drawings of any special equipment required.

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6.5 Surveillance sample. A five pound sample of propellant shall be forwarded to US Army Armament Research, Development and Engineering Center (ARDEC), Picatinny Arsenal, NJ 07806-5000, ATTN: SMCAR-AEE-WE for the propellant surveillance test program in accordance with MIL-STD-652.

6.6 Technical Agency. All correspondence to the technical agency shall be addressed to U.S. Army ARDEC, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, N.J. 07806-5000.

6.7 Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Throughout production, if 3 consecutive lots fail on first test or a lapse in production of 90 days or more occurs, testing shall revert back to initial production plan.

6.8 Subject term (key word) listing.

Ammunition
Ignition cartridge propellant
Mortar ammunition
Propellant

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1376-A329)

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